



UNIVERSITY
of HAWAII®
MĀNOA

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Summary

Postdoctoral Research Scientist developing statistical and machine learning methods for scientific modeling, prediction, and inference. Research spans sparse regression, statistical inference, scientific machine learning, and data assimilation, with experience developing open-source software and collaborating across disciplines.

Skills

- **Statistics & Data Science:** Regression, statistical inference, sparse modeling, Bayesian methods, clustering, time-series analysis
- **Scientific Machine Learning:** Physics-informed neural networks (PINNs), differential equations, automatic differentiation, deep learning, optimization
- **Data Assimilation & Numerical Modeling:** 4D-Variational Data Assimilation (4D-Var), dynamical systems, coupled physical-biogeochemical modeling
- **Programming & Scientific Computing:** Python (PyTorch, NumPy, SciPy), R, MATLAB, SQL, High-Performance Computing (HPC)
- **Languages:** Conversational Spanish

Academic Appointments

Postdoctoral Research Scientist

Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, HI

April 2024 – Present

Concurrent appointment: Research Affiliate, with PI eligibility (2026 – Present)

- Developed a physics-informed neural network framework for data assimilation that uses automatic differentiation to replace traditional model components with a single differentiable system, achieving a 6.5-fold computational speedup while maintaining comparable state accuracy.
- Validated learned dynamics through sensitivity analysis, demonstrating that physical constraints preserve realistic behavior while allowing the model to adapt to observations.
- Developing differentiable surrogate models to improve data assimilation and forecasting in the Regional Ocean Modeling System (ROMS), with extensions to coupled physical-biogeochemical models.

Education

PhD in Computational Engineering

Durham University, Durham, United Kingdom

October 2019 – December 2023

Dissertation: Automatic Extraction of Ordinary Differential Equations from Data: Sparse Regression Tools for System Identification

MSc in Data Science

Northwestern University, Evanston, IL

September 2017 – May 2019

BA in Statistics and Mathematics

Wittenberg University, Springfield, OH

August 2012 – May 2016

Peer-Reviewed Publications

Egan, K., Li, W. & Carvalho, R. Automatically discovering ordinary differential equations from data with sparse regression. *Commun Phys* 7, 20 (2024). <https://doi.org/10.1038/s42005-023-01516-2>

Preprints & Submitted Manuscripts

Egan, K., & Powell, B. Physics-Informed Neural Networks as Differentiable Surrogates for 4D-Variational Data Assimilation. Under review at the *Journal of Advances in Modeling Earth Systems (JAMES)*. ESS Open Archive preprint. <https://doi.org/10.22541/essoar.15008576/v1>

Research Software

- **ARGOS (Automatic Regression for Governing Equations):** Creator and maintainer of a CRAN-published R package for sparse system identification from time-series data. Implements methods introduced in "Automatically Discovering Ordinary Differential Equations from Data with Sparse Regression" (*Commun Phys*, 2024). [[CRAN](#) | [GitHub](#)]
- **PI-NPZ (Physics-Informed Neural Network for NPZ Dynamics):** Developed an open-source Python framework built with PyTorch for differentiable 4D-Variational Data Assimilation using automatic differentiation to unify forward, tangent-linear, and adjoint operators. Implements methods described in the submitted manuscript "Physics-Informed Neural Networks as Differentiable Surrogates for 4D-Variational Data Assimilation." [[GitHub](#) | [DOI](#)]

Teaching & Mentoring Experience

Undergraduate Research Mentor

University of Hawai‘i at Mānoa, Honolulu, HI

2026 – Present

- Mentor an undergraduate researcher developing machine learning methods for reconstructing subatomic particle tracks using the CERN TrackML dataset.

Workshop Instructor

CBIOMES Workshop on Machine Learning Methods for Time-Series Data

2026

- Taught the full second day on neural networks and physics-informed neural networks, developing lectures and hands-on exercises for researchers from varied disciplinary and technical backgrounds.

Language and Culture Assistant

IES Angel Correa, Madrid, Spain

September 2016 – May 2017

- Led classroom lessons in engineering subjects and developed instructional materials to present technical concepts more clearly and engagingly.

Math Workshop Tutor

Wittenberg University, Springfield, OH

January 2016 – May 2016

Elementary School Teaching Assistant

Wittenberg University, Springfield, OH

January 2015 – May 2015

Elementary Summer School Math Teacher

Susan H. Hagen Center, Springfield, OH

June 2014 – August 2014

Selected Presentations

Oral Presentations

- *Leveraging Physics-Informed Neural Networks for 4D-Variational Data Assimilation.* CBIOMES Annual Meeting, June 2025
- *Building a Physics-Informed Neural Network for NPZ Modeling.* CBIOMES Workshop on Photosynthesis and Primary Production, March 2025
- *Discovering Governing Equations from Data with Sparse Regression.* Joint Statistical Meetings (JSM) & Symposium on Data Science & Statistics (SDSS); SIAM Conference on Optimization (OP21) & Conference on Dynamical Systems (DS21), August 2021

Poster Presentations

- *Physics-Informed Neural Networks as Differentiable Surrogates for 4D-Variational Data Assimilation*. Machine Learning and Dynamical Systems (MLDS 5) & Differential Equations for Data Science (DEDS 2026), February 2026; Accelerating Research in the Age of AI (Google–UH), March 2026

Honors & Awards

Pacific Islands Impact Award

Accelerating Research in the Age of AI Workshop (Google–UH), 2026

Professional Leadership & Service

Lead Workshop Organizer

CBIOMES Workshop on Machine Learning Methods for Time-Series Data
2026

- Co-authored the successful Simons Foundation proposal and led the organization of a 2.5-day workshop on machine learning methods for time-series data.

Committee Member

Durham University, Postgraduate Research Conference
January 2022 – July 2022

Chairman, Delta Tau Delta

Wittenberg University, Springfield, OH
August 2013 – May 2016

Additional Professional Experience

Associate Business Analyst

Medline Industries, Northfield, IL
September 2017 – September 2019

Data and Assessment Assistant

Springfield Promise Neighborhood, Springfield, OH
January 2016 – May 2016

- Developed logistic regression models and data visualizations for program evaluation, contributing to successful funding acquisition.

Community Outreach

- STEM Night, Ho‘okele Elementary School — April 2026
- STEM Night, Mililani Middle School — March 2026
- Train of Hope, Vienna, Austria — September–December 2015

Research Affiliations

- Simons Collaboration on Computational Biogeochemical Modeling of Marine Ecosystems (CBIOMES), 2024–Present
- Hawai‘i Data Science Institute, 2024–Present
- Centre for Doctoral Training in Energy, Durham University, 2019–2023